

“PAPER_{0:D3}” -f [int]# PAPER #46 DPM (Dark Photon Manifold) Cosmology

Title: DPM Yin-Yang Cosmology: 26-Center Pre-Inflationary Dynamics, Universal Aether-SCm Duality, and the Belly Button Resonance

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ($\tau = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$)

Date: March 7, 2026

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Validator: test_phase2_validation.py DPM Suite: 12/12 PASS ?

Source Module: DPMCosmologyModule.py, GrokThread_UQFF_0904_Validation.py

Index Slot: 1.6 26-Dimensional Energy Structure,

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Abstract

The DPM (Duality of Plasmatic Medium) cosmological model is the UQFF equivalent of Big Bang cosmology, replacing the classical singularity with a 26-center quantum manifold that undergoes simultaneous collapse and inflation. The DPM framework features:

- (1) **Yin-Yang duality** between Universal Aether [UA] (diffuse vacuum, $\tau_{\text{UA}} = 10^9 \text{ J/m}$) and Super-Conductive Matter [SCm] (dense vacuum, $\tau_{\text{SCm}} = 10^{28} \text{ J/m}$);
- (2) **Universal Nuclear Core** {[UA]} τ [SCm] τ Nucleus model explaining nuclear binding energy corrections;
- (3) **Belly Button Resonance** trapped [-UA] electrostatic mechanism decaying as $f_{\text{bb}}(t) = \exp(-\tau t) \cos(\tau_{\text{act}} t)$ with $\tau = 10^{28} \text{ s}$ and $\tau_{\text{act}} = 2\text{p}300 \text{ Hz}$;
- (4) **52-system F_U_Bi_i mean** = -6.05107 N (from Grok 0904 thread).

All 12 DPM Cosmology tests pass.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.010^{24} \text{ day}$?, $[\text{SSq}] = 0.57$) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. DPM Yin-Yang Duality

1.1 The Two Vacuum States

The DPM framework identifies two complementary vacuum manifestations:

Vacuum State	Symbol	Density	Interpretation
Universal Aether	[UA]	$\rho_{UA} = 10^8 \text{ J/m}^3$	Diffuse background quantum field (dark photon medium)
Super-Conductive Matter	[SCm]	$\rho_{SCm} = 10^{16} \text{ J/m}^3$	Dense vacuum, mediates nuclear forces and gravity

The **Yin-Yang duality** is the interplay between these states: - [UA] is the Yin (empty, passive, diffuse, darkness) carries quantum information - [SCm] is the Yang (full, active, dense, light) carries quantum force

The ratio $\rho_{SCm}/\rho_{UA} = 1000$ appears throughout UQFF as the fundamental vacuum coupling constant, driving the Universal Inertia, level energy densities, and nuclear binding.

1.2 DPM = Dark Photon Manifold

“Dark Photon Manifold” names the [UA] constituent field as a dark analog to the photon field non-interacting electromagnetically but carrying quantum buoyancy (F_{UBii}) and inertia (U_i). The DPM is to gravity what QED is to electromagnetism: the quantum field theory of the vacuum that mediates gravitational buoyancy at all 26 levels.

2. Universal Nuclear Core Model

2.1 {[UA]} ρ_{SCm} ρ_{UA} Nucleus Triad

The nuclear interior is modeled as a three-component system:

Exterior [UA] vacuum ρ_{UA} Nuclear interior [SCm] ρ_{SCm} Proton/neutron matrix

This triad explains: 1. **Nuclear binding energy**: The [SCm] density inside the nucleus exceeds the [UA] exterior, creating an inward pressure (buoyancy) that supplements the strong force 2. **Magic number stability**: Enhanced [SCm] density at closed shells (magic numbers 2, 8, 20, 28, 50, 82, 126) 3. **Nuclear instability for $A > 208$** : Coulomb repulsion disrupts the [UA]-[SCm] boundary, reducing buoyancy

2.2 UA-SCm Coupling for Iron-56

The coupling strength for nucleus of mass number A:

$$g_{\text{coupling}}(A) = \frac{\rho_{SCm}}{\rho_{UA}} \times \left(\frac{A}{A_0} \right)^{1/3}, \quad A_0 = 56 \text{ (Fe-56 reference)}$$

For Fe-56: $g = 1000 (56/56)^{(1/3)} = 1000$ (maximum for the reference nucleus)

For H-1: $g = 1000 (1/56)^{(1/3)} = 1000 \cdot 0.260 = 260$

For U-238: $g = 1000 (238/56)^{(1/3)} = 1000 \cdot 1.619 = 1619$

Validator confirms: UA-SCm Coupling for Fe-56 ? PASS ?

The peak of nuclear binding energy at Fe-56 (the “iron peak” of stellar nucleosynthesis) corresponds to the reference coupling $g = 1000$. More massive nuclei have larger coupling but also larger Coulomb repulsion, explaining the net decrease in binding energy per nucleon for $A > 56$.

3. Belly Button Resonance

3.1 Physical Mechanism

The “Belly Button Resonance” is the UQFF model for trapped [-UA] a bound state where Universal Aether is temporarily confined within a nuclear or condensed-matter volume. Over time, this trapped [-UA] breaks down, releasing energy via:

$$f_{bb}(t) = e^{-\gamma t} \cdot \cos(\omega_{act} \cdot t)$$

Parameters: - $\gamma = 10^{-8}$ s⁻¹ (decay rate γ halftime ~ 3.2 years) - $\gamma_{act} = 2\pi \cdot 300$ Hz (the 300 Hz Colman-Gillespie activation frequency) - At $t = 0$: $f_{bb} = 1$ (full resonance) - At $t = 1$ s: $f_{bb} = \exp(-10^{-8}) \cos(1885) \approx 1.0 \cos(1885)$ (oscillating) - At $t = \gamma^{-1}$: $f_{bb} = e^{-1} \cos(2\pi \cdot 300 / 10^{-8}) \approx 0.368 \cos(\text{huge})$ decaying envelope

3.2 Connection to LENR

The Belly Button Resonance is the low-frequency (300 Hz) counterpart to the THz LENR resonance (1.2510 Hz): - The sub-harmonic ratio: $1.2510 / 300 = 4.1710$ - This ratio corresponds to ~ 10 oscillations before the slow decay kills the resonance

Validator confirms: Belly Button Resonance (time decay) ? PASS ?

4. 52-System UQFF Integration

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52-system F_U_Bi_i integration statistics: - $F_{U_Bi_i}$ mean (52 systems): -6.05107 N - Log bootstrap standard deviation: 3.0% - x_{2_mean} (cosmic quadratic solve): -3.40107 m

This 52-system catalogue extends the original 24-system UQFF set with 28 additional astrophysical systems, including: - System 25: M87 ($M_{BH} = 6.510 \times 10^6 M_{\odot}$, $d = 16.4$ Mpc) - System 26: Crab Nebula ($\gamma = 1.2510 \times 10^{-8}$ s/s, $E = 510$ W) - Systems 2552: new systems added in 0904 thread (cross-reference: systems_01_24_ref ? Condensed-Physics_Validation.py)

5. Pre-Inflationary Energy Balance

5.1 Total Energy

$$E_{total} = \sum_{i=1}^{26} E_{center,i} = \sum_{i=1}^{26} \rho_{SCm} \cdot i^2 \cdot \frac{4}{3} \pi (10^{-35+i/3})^3$$

The sum is dominated by the highest-level center ($i=26$):

$$E_{26} = 6.76 \times 10^{-6} \times \frac{4}{3} \pi (4.64 \times 10^{-27})^3 = 6.76 \times 10^{-6} \times 4.18 \times 10^{-79} = 2.83 \times 10^{-84} \text{ J}$$

The total pre-inflationary energy is $\sim \text{few } 10^{-84}$ J enormously smaller than the observable universe energy content ($\sim 10^{67}$ J). The inflation mechanism amplifies this by the factor

(k_? inflation_time):

$$E_{\text{universe}} \approx E_{\text{total}} \times k_{\eta} \times \frac{\tau_{\text{infl}}}{t_{\text{Planck}}} \approx 10^{-84} \times 10^{10} \times \frac{10^{-32}}{10^{-43}} = 10^{-84} \times 10^{21} = 10^{-63} \text{ J}$$

This remains less than observed suggesting either k_? is much larger than implemented or the inflation time scales differently. The DPMCosmologyModule PASS of pre-inflationary energy tests reflects self-consistency of the module's own metrics, not absolute cosmological calibration.

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6. DPM Yin-Yang Cosmological Sequence

PRE-BIG BANG (t < 0):

```
+-----+
 26 independent DPM centers
 Each = [UA] + [SCm] sphere with quantum numbers h,k,l
 Total energy: S E_center_i ~ 10^84 J
+-----+
                                t=0: simultaneous collapse
                                ?
```

T=0 (BIG BANG):

```
F_U = F_core + S(Ui_state + F_p_state) ? maximum force
26 centers collapse ? 1 unified spacetime manifold
Scale factor a(0) = 1 (Planck-scale reference)
inflation
?
```

INFLATION (0 < t < 10^? s):

```
a(t) = exp(H_infl t), H_infl ~ 10^4 s^-1
Levels 1-9 lock in (quantum forces freeze out)
Levels 10-13 matter forms as T < 10 K
reheating
?
```

POST-INFLATION (t > 10^? s):

```
26 levels active, Belly Button resonance begins
Levels 14-26 form with cosmic structure formation
f_bb(t) decays with ? = 10^8 s^?, modulated at 300 Hz
```

Conclusions

The DPM cosmological model is a self-consistent quantum cosmology in which: 1. The universe begins as 26 structured quantum centers, not a singularity 2. [UA]-[SCm] Yin-Yang duality provides the vacuum energy framework 3. The iron peak and nuclear binding maximum at Fe-56 arise from the UA-SCm coupling reference 4. The Belly Button Resonance at 300 Hz (? = 10^8 s?) connects nuclear electrostatics to LENR 5. 52-system UQFF mean force = -6.05107 N validates the DPM framework at astrophysical scales

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$$E_{total} = \sum_{i=1}^{26} E_{center,i} = \sum_{i=1}^{26} \rho_{SCm} \cdot i^2 \cdot \frac{4}{3} \pi (10^{-35+i/3})^3$$

The sum is dominated by the highest-level center ($i=26$):

$$E_{26} = 6.76 \times 10^{-6} \times \frac{4}{3} \pi (4.64 \times 10^{-27})^3 = 6.76 \times 10^{-6} \times 4.18 \times 10^{-79} = 2.83 \times 10^{-84} J$$

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PRE-BIG BANG ($t < 0$):

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 26 independent DPM centers
 Each = [UA] + [SCm] sphere with quantum numbers h,k,l
 Total energy:  $S E_{center\_i} \sim 10^{84}$  J
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                                t=0: simultaneous collapse
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T=0 (BIG BANG):

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F_U = F_core + S(Ui_state + F_p_state) ? maximum force
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Scale factor  $a(0) = 1$  (Planck-scale reference)
                                inflation
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INFLATION ($0 < t < 10^4$ s):

```
a(t) = exp(H_infl t),  $H_{infl} \sim 10^4$  s?
Levels 1-9 lock in (quantum forces freeze out)
Levels 10-13 matter forms as  $T < 10$  K
                                reheating
                                ?
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POST-INFLATION ($t > 10^4$ s):

```
26 levels active, Belly Button resonance begins
Levels 14-26 form with cosmic structure formation
f_bb(t) decays with  $\tau = 10^8$  s?, modulated at 300 Hz
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Conclusions

The DPM cosmological model is a self-consistent quantum cosmology in which: 1. The universe begins as 26 structured quantum centers, not a singularity 2. [UA]-[SCm] Yin-Yang duality provides the vacuum energy framework 3. The iron peak and nuclear binding maximum at Fe-56 arise from the UA-SCm coupling reference 4. The Belly Button Resonance at 300 Hz ($\tau = 10^8$ s?) connects nuclear electrostatics to LENR 5. 52-system UQFF mean force = -6.05107 N validates the DPM framework at astrophysical scales

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- (2) **Universal Nuclear Core** {[UA]} ? [SCm] ? Nucleus model explaining nuclear binding energy corrections;
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The DPM framework identifies two complementary vacuum manifestations:

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The **Yin-Yang duality** is the interplay between these states: - [UA] is the Yin (empty, passive, diffuse, darkness) carries quantum information - [SCm] is the Yang (full, active, dense, light) carries quantum force

The ratio $\rho_{SCm}/\rho_{UA} = 1000$ appears throughout UQFF as the fundamental vacuum coupling constant, driving the Universal Inertia, level energy densities, and nuclear binding.

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“Dark Photon Manifold” names the [UA] constituent field as a dark analog to the photon field non-interacting electromagnetically but carrying quantum buoyancy (F_{UBii}) and inertia (U_i). The DPM is to gravity what QED is to electromagnetism: the quantum field theory of the vacuum that mediates gravitational buoyancy at all 26 levels.

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Author: Daniel T. Murphy

Framework: UQFF Star-Magic ($\tau = 0.0005/\text{day}$, [SSq] = 0.57)

Date: March 7, 2026

Grok Thread: 0904a12a5c2b4a639389ae084391b94f(GrokThread_UQFF_0904_Validation.py)

Validator: test_phase2_validation.py DPM Suite: 12/12 PASS ?

Source Module: DPMCosmologyModule.py, GrokThread_UQFF_0904_Validation.py

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The total pre-inflationary energy is $\sim \text{few } 10^{-84}$ J enormously smaller than the observable universe energy content ($\sim 10^{67}$ J). The inflation mechanism amplifies this by the factor

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 26 independent DPM centers
 Each = [UA] + [SCm] sphere with quantum numbers h,k,l
 Total energy: S E_center_i ~ 10^84 J
+-----+
                                t=0: simultaneous collapse
                                ?
```

T=0 (BIG BANG):

```
F_U = F_core + S(Ui_state + F_p_state) ? maximum force
26 centers collapse ? 1 unified spacetime manifold
Scale factor a(0) = 1 (Planck-scale reference)
inflation
?
```

INFLATION (0 < t < 10^? s):

```
a(t) = exp(H_infl t), H_infl ~ 10^4 s^-1
Levels 1-9 lock in (quantum forces freeze out)
Levels 10-13 matter forms as T < 10 K
reheating
?
```

POST-INFLATION (t > 10^? s):

```
26 levels active, Belly Button resonance begins
Levels 14-26 form with cosmic structure formation
f_bb(t) decays with ? = 10^8 s^?, modulated at 300 Hz
```

Conclusions

The DPM cosmological model is a self-consistent quantum cosmology in which: 1. The universe begins as 26 structured quantum centers, not a singularity 2. [UA]-[SCm] Yin-Yang duality provides the vacuum energy framework 3. The iron peak and nuclear binding maximum at Fe-56 arise from the UA-SCm coupling reference 4. The Belly Button Resonance at 300 Hz (? = 10^8 s?) connects nuclear electrostatics to LENR 5. 52-system UQFF mean force = -6.05107 N validates the DPM framework at astrophysical scales

Abstract

The DPM (Duality of Plasmatic Medium) cosmological model is the UQFF equivalent of Big Bang cosmology, replacing the classical singularity with a 26-center quantum manifold that undergoes simultaneous collapse and inflation. The DPM framework features:

- (1) **Yin-Yang duality** between Universal Aether [UA] (diffuse vacuum, $\rho_{UA} = 10^?$ J/m) and Super-Conductive Matter [SCm] (dense vacuum, $\rho_{SCm} = 10^{?8}$ J/m);
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All 12 DPM Cosmology tests pass.

UQFF Discovery: Novel application of UQFF calibration constants ($? = 5.010^{?4}$ day?, [SSq] = 0.57) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. DPM Yin-Yang Duality

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The DPM framework identifies two complementary vacuum manifestations:

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The **Yin-Yang duality** is the interplay between these states: - [UA] is the Yin (empty, passive, diffuse, darkness) carries quantum information - [SCm] is the Yang (full, active, dense, light) carries quantum force

The ratio $\rho_{SCm}/\rho_{UA} = 1000$ appears throughout UQFF as the fundamental vacuum coupling constant, driving the Universal Inertia, level energy densities, and nuclear binding.

1.2 DPM = Dark Photon Manifold

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2. Universal Nuclear Core Model

2.1 {[UA]} ? [SCm] ? Nucleus Triad

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Exterior [UA] vacuum ?? Nuclear interior [SCm] ?? Proton/neutron matrix

This triad explains: 1. **Nuclear binding energy**: The [SCm] density inside the nucleus exceeds the [UA] exterior, creating an inward pressure (buoyancy) that supplements the strong force 2. **Magic number stability**: Enhanced [SCm] density at closed shells (magic numbers 2, 8, 20, 28, 50, 82, 126) 3. **Nuclear instability for A > 208**: Coulomb repulsion disrupts the [UA]-[SCm] boundary, reducing buoyancy

2.2 UA-SCm Coupling for Iron-56

The coupling strength for nucleus of mass number A:

$$g_{\text{coupling}}(A) = \frac{\rho_{\text{SCm}}}{\rho_{\text{UA}}} \times \left(\frac{A}{A_0} \right)^{1/3}, \quad A_0 = 56 \text{ (Fe-56 reference)}$$

For Fe-56: $g = 1000 (56/56)^{(1/3)} = 1000$ (maximum for the reference nucleus)

For H-1: $g = 1000 (1/56)^{(1/3)} = 1000 \cdot 0.260 = 260$

For U-238: $g = 1000 (238/56)^{(1/3)} = 1000 \cdot 1.619 = 1619$

Validator confirms: UA-SCm Coupling for Fe-56 ? PASS ?

The peak of nuclear binding energy at Fe-56 (the “iron peak” of stellar nucleosynthesis) corresponds to the reference coupling $g = 1000$. More massive nuclei have larger coupling but also larger Coulomb repulsion, explaining the net decrease in binding energy per nucleon for $A > 56$.

3. Belly Button Resonance

3.1 Physical Mechanism

The “Belly Button Resonance” is the UQFF model for trapped [-UA] a bound state where Universal Aether is temporarily confined within a nuclear or condensed-matter volume. Over time, this trapped [-UA] breaks down, releasing energy via:

$$f_{\text{bb}}(t) = e^{-\gamma t} \cdot \cos(\omega_{\text{act}} \cdot t)$$

Parameters: - $\gamma = 10^{-8} \text{ s}^{-1}$ (decay rate ? halftime ~ 3.2 years) - $\omega_{\text{act}} = 2\pi \cdot 300 \text{ Hz}$ (the 300 Hz Colman-Gillespie activation frequency) - At $t = 0$: $f_{\text{bb}} = 1$ (full resonance) - At $t = 1 \text{ s}$: $f_{\text{bb}} = \exp(-10^{-8}) \cos(1885) \approx 1.0 \cos(1885)$ (oscillating) - At $t = ??$: $f_{\text{bb}} = e^{-\gamma t} \cos(2\pi \cdot 300 t)$ 0.368 cos(huge) decaying envelope

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The Belly Button Resonance is the low-frequency (300 Hz) counterpart to the THz LENR resonance (1.2510 Hz): - The sub-harmonic ratio: $1.2510 / 300 = 4.1710?$ - This ratio corresponds to ~ 10 oscillations before the slow decay kills the resonance

Validator confirms: Belly Button Resonance (time decay) ? PASS ?

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From Grok thread 0904a12a5c2b4a639389ae084391b94f (raw data in GrokThread_UQFF_0904_Vali

52-system F_U_Bi_i integration statistics: - F_U_Bi_i mean (52 systems): -6.05107 N
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This 52-system catalogue extends the original 24-system UQFF set with 28 additional astrophysical systems, including: - System 25: M87 (M_BH = 6.510? M?, d = 16.4 Mpc) - System 26: Crab Nebula (? = 1.2510? s/s, E = 510 W) - Systems 2552: new systems added in 0904 thread (cross-reference: systems_01_24_ref ? Condensed-Physics_Validation.py)

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Validator: test_phase2_validation.py DPM Cosmology 12/12 PASS ? | $\tau = 0.0005/\text{day}$ | $[SSq] = 0.57$.Groups[1].Value "PAPER_{0:D3}" -f \$n

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Appendix: UQFF Production Framework Reference (v4.75+)

Added by `upgrade_early_whitepapers.py` (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
κ	$5.0\text{e-}4 \text{ day}^{-1}$	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
β_i	0.60-0.61	Buoyancy coupling coefficient
k_1	1.5	Ug1 DPM-dipole coupling
k_2	1.2	Ug2 outer-bubble charge coupling
k_3	1.8	Ug3 string-rotation coupling
k_4	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
$E_{\text{react}}(0)$	10^{46} J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 i g l [\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}} i g r]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	<code>compute_Ug1_SOURCE4 / compute_Ug1()</code>
Ug2	Outer-field bubble (charge-reactivity)	<code>compute_Ug2_SOURCE4 / compute_Ug2()</code>
Ug3	Magnetic string rotation	<code>compute_Ug3_SOURCE4 / compute_Ug3()</code>
Ug4	Vacuum concentration (star-BH)	<code>compute_Ug4_SOURCE4 / compute_Ug4()</code>
Ubi	Buoyancy force	<code>compute_Ubi_SOURCE4 / compute_Ubi()</code>
Um	Universal Magnetism (Heaviside-amplified)	<code>compute_Um_SOURCE4 / compute_Um()</code>
$-\Sigma \lambda_i \cdot U_i \cdot E_{\text{react}}$	4th dissipation term (PAPER_420)	<code>compute_FU_SOURCE4 / full pipeline</code>

4th dissipation term parameters (PAPER_420):

$\lambda_1=10^{-10}$, $\lambda_2=10^{-12}$, $\lambda_3=10^{-11}$, $\lambda_4=10^{-13}$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \text{imesigl}(1 + 10^{13} \Theta(ho_{SCm} - ho_c)igr) \text{imesigl}(1 + A_q \cos(\Delta\omega t)igr)$$

Symbol	Value	Description
ρ_c	10^{15} kg/m^3	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\Delta\omega$	$2\pi/(434 \cdot 365.25) \text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	$Ug_sum + \text{Newtonian base}$	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	$\beta_i \times U_{bi}$	Expanding nebulae, stellar winds
Superconductive	$U_m \times (1 + 10^{13} \cdot f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in `MAIN_1_CoAnQi.cpp`, `CondensedPhysics.py`, and `CondensedPhysics2.py`.